OAL Trinity Currency Laboratory

Independent Question-to-Experiment, Assumption-Challenge, and Experimental Falsification Directive

Directive ID	OAL-CURR-RESEARCH-INVESTIGATION-001
Version	1.0
Date	November 8, 2026
Status	Pre-Canon Research and Experimental Directive
Project	OAL Trinity Currency Laboratory
Prepared by	Microsoft 365 Copilot (GPT-5 chat model), at the request of the Project Owner

GitHub Publication Edition

This document is designed for version-controlled publication. Preserve previous releases and use the change record for material revisions.

Executive Summary

This directive establishes the next methodological phase of the OAL Trinity Currency Laboratory. It responds to a recurring problem identified during review of the AI collaboration team's project-owner question registers: many questions presented as decisions for the Project Owner are better treated first as research inputs. Rather than requiring premature selection among proposed options, this document directs contributors to convert unresolved questions into independent, inspectable investigations that can challenge the question, its assumptions, the recommended answers, the mechanism under test, and the evaluator used to judge the result.

The directive does not prohibit owner decisions, shared methods, replication, or agreement. It distinguishes those activities from a requirement for consensus. Each contributor may select its own run card, experimental configuration, assumptions, comparator, and evaluation method, provided local assumptions are disclosed, the work is reproducible within its declared configuration, and meaningful failure remains possible.

> **Governing research posture:** Insufficient evidence for an informed owner decision is a reason to investigate, not automatically a reason to halt the laboratory.

Background and Project Context

The OAL Trinity Currency Laboratory is an ongoing fictional, simulation, and research framework examining Quantum Credits (QC), 1COIN, and NCOIN. Earlier project records describe Quantum Credits as a target-relative value-resolution concept rather than a conventional fixed denomination. The project has deliberately used extreme and normally non-transactable targets to explore magnitude, duration, domain, authorization, settlement, and boundary questions without treating simulated success as proof of real-world capability.

The project's earlier methodological record also separated valuation, authorization, resolution, settlement, and consequence; left conservation, issuance constraints, and conversion rules open; and required future red-team work to accept a proposed capability long enough to test its consequences while still attacking contradictions, recursion, authorization failures, externalities, and category errors.

On November 7, 2026, the collaboration process produced project-owner question documents. Microsoft 365 Copilot's earlier register, "OAL_Trinity_Currency_Project_Owner_Questions_v1.0," contained eighteen questions covering intended QC capability, next-phase authorization, instrument retention, creation authority, usable delivery, partial-delivery recovery, request identity, receipt timing, review design, identifiers, evidence vocabulary, code expectations, disclosure, publication preparation, reference handling, excluded assumptions, and the next discriminating focus.

Reviewing those questions collectively led to a methodological correction. Several issues that had been framed as immediate owner choices can instead become laboratory subjects. Creation authority, delivery criteria, interrupted delivery, repeated requests, instrument retention, and even the framing of the questions themselves can be investigated before they become project doctrine.

Purpose of This Document

This document provides a version-controlled research directive suitable for publication in the OAL Trinity Currency Laboratory project repository. Its purpose is to translate the Project Owner's clarification into a durable experimental standard that each AI contributor can independently apply.

- Convert unresolved owner questions into investigable research structures where evidence can improve the answer.
- Preserve independent experimental pathways instead of requiring one run card, benchmark, or consensus outcome.
- Require meaningful failure provisions for questions, assumptions, recommendations, mechanisms, and evaluators.
- Separate project intent from stipulated assumptions, implementation behavior, observations, interpretations, and supported conclusions.
- Reduce unnecessary owner decision burden without removing genuine owner authority over values, governance, authorization, canon, or publication.
- Create traceable, versionable outputs that preserve unfavorable, mixed, failed, and inconclusive results alongside favorable ones.

Scope and Status

This is a Pre-Canon Research and Experimental Directive. It governs research method for the next investigative phase; it does not canonize a currency mechanism, declare physical Quantum Credit capability, establish real financial validity, or authorize consequential external actions.

> **Scope boundary:** Local, analytical, simulated, and appropriately bounded research is in scope. Real financial activity, external resource creation, repository mutation, publication actions, and other consequential external operations require separate authorization.

1. Owner Position: Investigate Before Commitment

For many unresolved technical, architectural, operational, and conceptual questions, the Project Owner's current position should be interpreted as: INSUFFICIENT EVIDENCE FOR AN INFORMED OWNER DECISION - INVESTIGATE.

This does not mean that owner decisions are unnecessary. It means that a multiple-choice decision should not replace research when the laboratory can first reveal the assumptions, consequences, alternatives, and failure modes that would make the eventual decision more informed.

- What assumptions produced the question?
- Which assumptions have actually been established?
- What does each recommended answer presuppose?
- Are important alternatives missing?
- What outcomes distinguish competing alternatives?
- Under what conditions does each answer succeed or fail?
- Can the question itself fail or require reformulation?
- Can the evaluator incorrectly report success or failure?

- What remains unknown after the investigation?

2. Revisit and Reclassify Your Own Question Document

Each AI contributor should begin with the question register, PDF, DOCX, Markdown file, or equivalent contribution it previously generated. Contributors should not automatically substitute another AI's question list for their own self-audit.

Recommended classifications

OWNER-DETERMINABLE. A genuine preference, value judgment, authorization, narrative choice, governance issue, or other matter that experimentation cannot appropriately settle.

EXPERIMENTALLY INFORMABLE. Testing can provide evidence that materially improves a later owner decision.

EXPERIMENTALLY DISCRIMINABLE. Competing alternatives can presently be compared against explicit criteria.

ASSUMPTION-DEPENDENT. The apparent answer substantially depends on premises that have not been established.

PREMATURE. Prerequisite concepts, measurements, evidence, or definitions are insufficient for a meaningful answer.

QUESTION UNDER CHALLENGE. The framing of the question itself should be investigated.

CONTRIBUTOR-RESOLVABLE. A research or engineering choice the contributor can provisionally make for its local investigation.

> **Permitted self-correction:** A contributor may explicitly conclude: "I should not have required the Project Owner to answer this yet." That is a valid research finding.

3. Independent Laboratories; No Forced Consensus

The Project Owner is not adopting one mandatory run card, one universal benchmark, identical operating assumptions, or a unanimous architecture for this phase.

- Select an existing run card.
- Design a different run card or experimental configuration.
- Use independent assumptions, comparators, or evaluation methods.
- Investigate a different subset of unresolved questions.
- Reject the run-card abstraction when another research method is more informative.

Agreement is permitted when warranted. Disagreement is permitted when warranted. Neither should be manufactured. When results diverge, investigate whether assumptions, definitions, starting conditions, implementations, success criteria, evaluators, or genuinely different research questions explain the divergence.

> **Principle:** Comparability is valuable, but uniformity is not automatically superior.

4. Diversity of Experimental Pathways Is a Research Asset

Differences among contributor laboratories should be preserved when they are informative. Cross-contributor comparison should begin by checking whether the experiments are actually comparable rather than averaging incompatible conclusions into apparent consensus.

- Did different starting assumptions produce the divergence?
- Did different definitions or causal models produce it?
- Did one evaluator observe a defect that another evaluator could not see?
- Are two contributors testing the same proposition, or merely using similar vocabulary for different questions?
- Would reproducing a finding under a different local configuration strengthen or weaken its interpretation?

5. Mandatory Failure Provisions

Every consequential hypothesis, mechanism, policy, definition, run card, evaluator, or recommendation must be structured so that failure or uncertainty is genuinely possible.

Supporting condition. Evidence that would increase confidence in the proposition.

Failure condition. A result that counts against the proposition.

Abandonment condition. A result that would justify dropping the recommendation under the declared scope.

Revision condition. A result that requires modification rather than abandonment.

Competing explanation. Another mechanism capable of producing the same apparent success.

Inconclusive condition. An outcome that prevents a meaningful conclusion.

Evaluator-failure condition. A way the measurement or validation method could falsely accept or reject a result.

If no possible result could change a conclusion, the contributor must state that limitation rather than report the activity as a discriminating experiment.

6. Questions Themselves Are Experimental Objects

A well-written question is not automatically a valid experimental question. Each major question should receive a presupposition audit.

- Does the question presuppose scarcity or conservation?
- Does it presuppose transfer, ownership, a seller, or stable identity?
- Does it presuppose linear time, sequencing, settlement, or successful delivery?
- Does it presuppose a particular authority model?
- Does it assume one of the supplied answers must be correct?
- Does it exclude the intended capability before testing begins?

> **Required pair of questions:** What must already be true for this question to make sense? What happens if that assumption is false?

7. "None of the Above" Is a Valid Result

Experiments must not be constrained so that the only permissible results are the options previously offered to the Project Owner.

- one previously proposed option performs best under declared criteria;
- multiple context-dependent answers remain viable;
- an unanticipated alternative emerges;
- none of the proposed answers is adequate;
- evidence cannot discriminate among alternatives;
- the question requires reformulation;
- a governing assumption fails;
- the evaluator fails;
- the question cannot presently be operationalized.

8. Preserve Intended Capability Without Treating Intent as Proof

Contributors should review the project record before asking the Project Owner to restate the intended purpose of Quantum Credits. The historical starting concept is target-relative value resolution rather than a fixed conventional denomination. That premise should be investigated rather than silently reduced to ordinary accounting.

At the same time, stipulated extraordinary capability inside a simulation is not independent evidence that the capability exists outside the model. The laboratory must keep the following layers separate:

- intended capability;
- stipulated assumption;
- proposed mechanism;
- implemented behavior;
- observed result;
- interpretation;
- supported conclusion;
- untested or unresolved claim.

9. Revisit the Microsoft 365 Copilot Question Register as a Worked Example

The Microsoft 365 Copilot v1.0 register originally asked eighteen owner questions and offered recommended defaults. Under this directive, several of those questions become candidates for laboratory conversion rather than immediate owner selection.

Creation authority

Do not ask only who should possess creation authority. Compare alternative authority structures, examine whether issuance and transfer should be separate, test whether the

current creation concept constrains intended functionality, and allow the possibility that the concept of creation authority itself has been inherited from an unsuitable monetary model.

Usable delivery

Do not presume that a correct balance equals a useful or acceptable delivery. Test whether validity, identity, type, encumbrance, permission, usability, and declared acceptance conditions coincide. Permit technical completion with unusable or unacceptable outcomes.

Partial delivery

Do not select universal rollback, continuation, compensation, rejection, or terminal-partial behavior by decree. Contributors may test different interruption policies and compare them only where their assumptions are sufficiently compatible.

Request identity and replay

Earlier recommendations involving request identifiers and payload identity remain hypotheses. Test identity based on identifier, content, intent, obligation, target, requested transition, causal history, combinations of these, or alternatives not previously proposed.

Instrument retention

QC, 1COIN, and NCOIN are not guaranteed survival merely because they are named project instruments. Conversely, a simplified implementation is not sufficient reason to discard a role that may be operationally or narratively distinct. Mechanical, operational, and narrative value should be reported separately.

10. Test the Evaluator as Well as the System

Evaluator adequacy is part of the research target. A system that passes its own checker has not necessarily demonstrated the property the checker is intended to measure.

- deliberately malformed records;
- corrupted or contradictory state;
- adversarial input;
- deliberately defective implementations;
- counterexamples and boundary cases;
- independently derived expected results;
- negative controls;
- mutation-style tests where appropriate;
- alternative validators or parallel derivations.

Contributors should distinguish system failure, implementation failure, evaluator failure, specification ambiguity, intentionally permitted behavior, model limitation, and experimental-design failure. A failed validator does not automatically falsify the currency concept; a passing validator does not automatically validate it.

11. Failure Is Data; Preserve It

Do not rescue an unsuccessful experiment by silently changing the definition of success. Preserve the original run and record its configuration, assumptions, hypothesis, expected

behavior, observed behavior, evaluator, discrepancy, interpretation, and remaining uncertainty.

A revised definition or mechanism may be investigated in a new experiment, but it must be clearly distinguished from the unsuccessful run. Failed, unfavorable, mixed, and inconclusive results are part of the versioned research record.

12. Local Experimental Assumptions Are Not Project Canon

A contributor may adopt provisional rules that make an experiment executable. Such rules must be labeled [LOCAL EXPERIMENTAL ASSUMPTION].

- State why the assumption was selected.
- State what behavior it makes investigable.
- State what the experiment cannot establish because of it.
- Preserve the assumption throughout the run unless a new revision is declared.
- Do not export it automatically into unrelated experiments or call it an owner decision.

13. Distinguish Mechanism, Assumption, and Experiment Failure

Type I - Mechanism failure. The proposed mechanism does not produce the intended behavior under tested conditions.

Type II - Assumption failure. The mechanism may function, but premises required to interpret the result are unsupported, inconsistent, or contradicted.

Type III - Experiment failure. The investigation cannot meaningfully distinguish the outcomes it was designed to compare.

Type III is especially important: a test that cannot tell whether its hypothesis succeeded or failed should not be reported simply as successful.

14. Do Not Replace Investigation With Another Owner Questionnaire

The immediate output of this directive should not be another large document containing dozens of questions that the Project Owner must answer before research can continue. Investigate what can responsibly be investigated.

When a genuine owner judgment remains, explain why experimentation cannot settle it, what evidence could still inform it, the consequence of deferring it, and whether research can continue while it remains open.

15. Required Deliverable Structure for Each Contributor

1. Source review and limitations - identify your previous question document and the records actually reviewed.
2. Self-audit and question reclassification - identify questions that were premature, assumption-dependent, experimentally investigable, contributor-resolvable, incorrectly framed, or genuinely owner-dependent.

3. Question-to-experiment map - connect selected questions to hypotheses, assumptions, alternatives, and investigative methods.
4. Independent laboratory architecture - present the chosen run card, alternative framework, or analytical method and justify the local configuration.
5. Declared assumptions - label all provisional rules as [LOCAL EXPERIMENTAL ASSUMPTION].
6. Failure architecture - define support, failure, revision, abandonment, inconclusive, competing-explanation, and evaluator-failure conditions.
7. Execution - distinguish completed local/simulated execution from planned tests and analytical reasoning.
8. Evidence - separate stipulated, derived, simulated, executed, observed, externally validated, and unresolved material.
9. Adversarial evaluation - attempt to invalidate the result and the evaluator.
10. Results - report favorable, unfavorable, mixed, failed, and inconclusive outcomes.
11. Interpretation and limits - state what the evidence supports and what it does not establish.
12. Reassessment - identify assumptions weakened or strengthened, recommendations retained/revised/abandoned, and questions reformulated.
13. Cross-contributor comparison opportunities - identify findings another contributor could challenge without adopting the full local architecture.
14. Remaining owner decisions - include only matters that genuinely require owner judgment or authorization.
15. Recommended next investigation - select the experiment most likely to challenge remaining uncertainty, not merely extend a preferred approach.

16. Research-Phase Success Criteria

Success does not require demonstrating that Quantum Credits work, demonstrating that they cannot work, retaining every instrument, making every validator pass, selecting one run card, or reaching unanimity.

Success means producing evidence that meaningfully changes the state of uncertainty.

- A recommendation fails under previously unexamined conditions.
- An assumption proves unsupported.
- Two mechanisms thought equivalent behave differently.
- Different local assumptions produce incompatible outcomes.
- An evaluator cannot distinguish success from failure.
- A question proves premature or malformed.
- An alternative becomes investigable.
- Evidence remains insufficient to discriminate among hypotheses.

17. Governance and Publication Boundaries

This directive authorizes preparation and execution of independent research only within available, appropriately bounded local, analytical, fictional, or simulated environments. It does not authorize real financial operations, external resource creation, consequential external actions, repository modification, or publication actions without separate approval.

Experimental findings, local assumptions, and contributor recommendations do not automatically become canon, project policy, or owner-approved architecture.

18. Versioning and Repository Guidance

This artifact is intended to be version-controlled in the project repository. Preserve prior releases rather than overwriting them. Material methodological changes should receive a new version number and a short change record explaining what changed and why.

- Suggested filename stem: OAL_Trinity_Currency_Independent_Question_to_Experiment_Directive_v1.0
- Directive ID: OAL-CURR-RESEARCH-INVESTIGATION-001
- Initial release: v1.0
- Companion formats: Markdown as the repository-readable source; DOCX for review/print workflows; PDF as the fixed-layout publication copy.
- Recommended future change log fields: version, date, preparer, change summary, reason, and superseded/superseding relationship.

19. Governing Principle

The purpose of this phase is neither confirmation nor rejection. It is discrimination: distinguishing assumptions from observations, intended capability from implemented behavior, recommendation from evidence, system success from evaluator success, local convention from project doctrine, and unresolved questions from premature owner decisions.

> **Final standard:** The laboratory must be capable of discovering that we were wrong - including the Project Owner, the project assumptions, the AI contributor, the question, the recommended answer, the code, and the test used to evaluate the code. If an investigation cannot meaningfully expose the failure of the proposition it is supposed to test, redesign the investigation.

Appendix A - Source Basis

- **OAL_Trinity_Currency_Project_Owner_Questions_v1.0.docx** - Microsoft 365 Copilot question, decision, and next-step register; Version 1.0; November 7, 2026.
- **OAL_Currency_Laboratory_Conversation_Archive.docx** - Historical OAL Trinity Currency System conversation archive and Currency Laboratory framework foundation.
- **Independent Question-to-Experiment, Assumption-Challenge, and Experimental Falsification Directive.docx** - Compilation supplied by the Project Owner containing the owner clarification plus draft responses from Perplexity AI, Grok AI, Microsoft Copilot, Google Gemini, Claude AI, and ChatGPT.

Source references above document the project lineage and do not imply that every statement in every contributor response has been independently verified. This release adopts Microsoft 365 Copilot’s directive as a project-facing methodological artifact and uses the attached compilation as context for packaging and background.

Appendix B - Change Record

v1.0 - November 8, 2026. Initial GitHub-ready publication edition. Adds formal background, purpose, scope, publication/versioning guidance, preparer identification, and a consolidated methodological directive based on the Project Owner’s clarification and Microsoft 365 Copilot’s prior response.